Rank-Fidelity-Gated Optimization: Keeping Certificate Descent Safe for Learned System Components

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Abstract

Learned components now ship inside systems with formal certificates: algebraic-multigrid (AMG) prolongators, neural Lyapunov controllers, interval-bound-propagation (IBP) certified classifiers. The operating assumption is that minimizing the certificate also minimizes the true objective. We build the missing check into that loop. We construct the rank-fidelity monitor, the Spearman correlation r_S between a certificate c and its truth t over the family $\mathcal{F}$ an operator intends to optimize, and we gate every certificate-directed descent: descend only when $r_S > 0.9$, otherwise restrict to discrete argmin over few candidates. We calibrate and falsify the gate in AMG, where bound, truth, and landscape are exactly computable: a per-edge prolongator lowers the certificate 3.1% while degrading the solver *3.1*× (85 to 262 PCG iterations), the monitor flags that family and accepts the faithful one-parameter one ($r_S = 0.974$). The gate reproduces its verdicts in Lyapunov synthesis and IBP robustness. Every number regenerates from committed artifacts in the public repository.

1 Introduction

A central promise of certified computation is that optimizing against a formal bound produces systems that are both performant and provably safe. That promise now underpins the three ways machine learning touches deployed systems: *learned numerical components* (neural preconditioners and smoothers inside large-scale solvers), *learned controllers* (neural Lyapunov functions certifying closed-loop stability), and *robust models* (IBP-certified classifiers). In each line, training against the certificate treats it as a faithful surrogate.

The gap: verification checks that a certificate is *valid*, never that descent against it is *safe*. Validity says the bound holds at a point; safe descent needs it to rank candidates along the path an optimizer walks. These are different quantities, and the second has no computable prerequisite. Optimizers find slack: a valid-but-vacuous certificate can be lowered arbitrarily without moving the objective. We close that gap with a method rather than a warning.

We build a *rank-fidelity monitor* that measures, over the family $\mathcal{F}$ an operator intends to optimize, the Spearman rank correlation r_S between the certificate c and the ground truth t (Sec. 3), and a *fidelity gate* around every certificate-directed descent: descend when $r_S > 0.9$, otherwise restrict to a discrete argmin over few candidates. We construct and falsify the gate in the one exactly computable setting, AMG preconditioner selection (Sec. 2), and we show the monitor's verdicts reproduce in neural Lyapunov synthesis and IBP-certified robustness (Sec. 4). We evaluate the gate end to end: it separates a faithful one-parameter AMG family ($r_S = 0.974$, descend) from an adversarial per-edge family ($r_S < 0.5$, restrict) that lowers the certificate 3.1% while degrading the solver 3.1×. We release code and committed per-seed records so every claim, including a negative CIFAR-10 scale run, is checkable end to end (Appendix). The position is falsifiable: validity and usefulness as an optimization target are independent, and their rank fidelity is the computable prerequisite for safe descent.

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2 The testbed: AMG preconditioner selection

We study the smoothed-aggregation two-level cycle for $-\nabla \cdot (K\nabla u) = f$ on 3D grids, with an exact coarse solve. For a prolongator P with aggregates T, the local spectral certificate is

$$K_T \;=\; \lambda_{\max}\big(A_T^{-1}\, R_T^{\top} \bar{M}_T R_T\big), \tag{1}$$

where A is the system matrix, A_T its principal submatrix on the patch T, $R_T = I - \Pi_T$ with Π_T the $\bar{M}$-orthogonal projection onto $\mathrm{range}(P_T)$ restricted to the patch, and $\bar{M} = D\,\sigma(D^{-1}A)$ the symmetrized smoother from the diagonal $D = \mathrm{diag}(A)$ and spectral radius $\sigma(\cdot)$. The two-level convergence factor $\hat{\rho}$ of the cycle satisfies the bound

$$\hat{\rho}^2 \;\le\; 1 - \frac{1}{N_{\mathrm{ov}}\, \max_T K_T}, \tag{2}$$

with N_{ov} the maximum aggregate overlap, so a smaller $\max_T K_T$ promises a faster solver. The certificate costs $\mathcal{O}(n)$ via batched Cholesky plus eigensolve on roughly 12×12 patches; at 32^3 unknowns it costs 2–4$\times$ a single PCG solve. The truth t is the PCG iteration count to $\|r\|/\|b\| < 10^{-8}$, equivalently the implied convergence factor ρ read off the iteration count. The baseline is the free strength-of-connection (SoC) heuristic, three numbers, the mean $|A_{ij}|$ per axis, standard in AMG since the 1980s [Ruge and St
"uben, 1987].

The testbed is exact in all three objects at once: the certificate (Eq. 1), the truth (measurable without error), and the optimization landscape (one weight per edge of the prolongator graph, or a single scaling ω). No learned benchmark we know of exposes this triple exactly, and that is what lets us separate "the bound is valid" from "descent against the bound is safe."

3 Method: the rank-fidelity monitor and the gate

> **Definition (rank fidelity).** For a family of candidates $\mathcal{F} = \{x(\theta)\}$ with certificate value $c(\theta)$ and truth $t(\theta)$, the rank fidelity is the Spearman rank correlation
>
> $$r_S(\mathcal{F}) \;=\; \rho_S\big(c(\theta), t(\theta)\big) \quad \text{over candidates } \theta \text{ sampled from } \mathcal{F},$$
>
> The slack is $s(\theta) = c(\theta) - t(\theta) \ge 0$.

The failure is elementary, so we state it as a proposition.

Proposition 1 (Validity does not imply descent safety)**.** *Let $c(\theta) \ge t(\theta)$ be a valid upper certificate for the truth t on a family Θ, with slack $s(\theta) = c(\theta) - t(\theta) \ge 0$. If s varies over a range exceeding that of t and is not positively correlated with t, then $\arg\min_{\theta\in\Theta} c(\theta)$ can be attained at a point where t is arbitrarily far from $\min t$, with c valid at every point.*

Proof. Take $\Theta = [-1, 1]$, $t = \theta^2$, $c = 1 + (\theta - 1)^2 \ge \theta^2$: $\min c$ at $\theta = 1$ where $t = 1$; $\min t = 0$ at $\theta = 0$ where $c = 2$, so descending c converges to the worst truth with c valid everywhere. □

The proof is deliberately elementary; the content is the measurement. Before any certificate-directed descent:

1. *Sample* the intended family $\mathcal{F}$: deterministic grids where structure allows (30-point AMG sweep, Lyapunov widths), randomly configured instances elsewhere (25-model IBP sweep, 120-model CIFAR manifest), the per-edge AMG family on its 8 held-out fields; then evaluate c and t on the sample.
2. *Measure* r_S. If $r_S > 0.9$, the certificate ranks the truth faithfully over this family, and descent is the safe regime of one-parameter families, conformal thresholds, and scalar calibration.
3. If $r_S \le 0.9$, *restrict* the search to a discrete argmin over few candidates, where no continuous slack exists.

The gate converts a faith-based practice into a testable hypothesis with a computable prerequisite. What is proved is Proposition 1; what is measured is the fidelity of every family in Table 1, and the gate's two claims, that it separates faithful from adversarial AMG families and that the same verdicts carry across the Lyapunov and IBP domains, are falsifiable (Sec. 4).

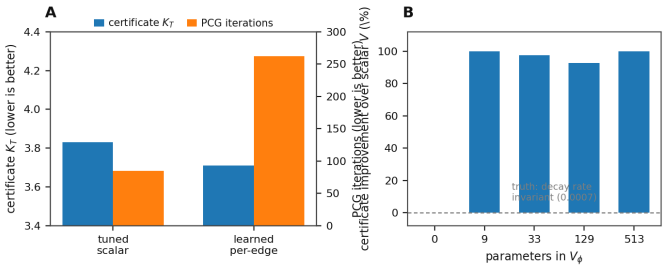


Figure 1: The failure the gate catches, in two domains (committed records). (A) AMG, jump class, $n = 8$ held-out fields, no cross validation, baseline the tuned scalar: the learned per-edge prolongator lowers the certificate K_T from 3.83 to 3.71 while PCG iterations to $\|r\|/\|b\| < 10^{-8}$ grow from 85 to 262, a 3.1% certificate gain bought at a 3.1× solver penalty. (B) Neural Lyapunov, Van der Pol: certificate improvement over the scalar $V = \|x\|^2$ baseline for each network width ($n = 5$ networks, one oscillator, matched families, no cross validation); the truth, the simulated decay rate 0.0007, is invariant. The rank fidelity of family (A)'s per-edge space is below the gate threshold; the one-parameter AMG control and the scalar Lyapunov baseline are above it.

4 Experiments: two results

4.1 Result 1: the gate separates faithful from adversarial descent in AMG

The question: same certificate, two families, opposite outcomes. The experiment: train a per-edge prolongator network, a 5-feature MLP producing smoothness weights $\Theta \in \mathbb{R}^{|E|}$ over the edge set E, against the local certificate with a log-sum-exp loss over patches, 18 epochs at $\mathrm{lr} = 3 \times 10^{-3}$, trained on 16 2D anisotropic-diffusion fields and tested on 8 held-out fields. In parallel, sweep the one-parameter family $P(\omega) = P_{\text{tent}} - \omega D^{-1} A P_{\text{tent}}$, with P_{tent} the tentative prolongator, over 30 scaling values and rank the same certificate against the same truth.

Tuned scalar: $K_T = 3.83$, $\rho = 0.8047$, 85 iterations
Learned per-edge: $K_T = 3.71$, $\rho = 0.9321$, 262 iterations

What came out: the learned prolongator lowered the certificate 3.1% while degrading the solver 3.1× (85 to 262 iterations), and the bound was never violated, $\hat{\rho}^2 \le 1 - 1/\max_T K_T$ holds for it too. The optimizer found where the certificate was slack, Goodhart's law [Goodhart, 1975] with exactly computable ground truth. The one-parameter family is the control: there the same certificate is faithful, Spearman $r_S = 0.974$ over the 30 values, certificate argmin ($\omega = 1.80\times$ textbook) within 5% of the truth's (1.90×).

What it means (Fig. 1A): the bound is not the failure mode, the *family* is.Over $\mathbb{R}^1$ the certificate tracks the truth; over $\mathbb{R}^{|E|}$ the optimizer finds slack directions. The gate measures this before descent: it accepts the one-parameter family ($r_S = 0.974 > 0.9$) and restricts the per-edge family ($r_S < 0.5 \le 0.9$) to discrete argmin, the regime where the certificate's picks are measured and correct (App. B, Fig. 3B).

4.2 Result 2: the same gate verdicts reproduce across control and robustness

Neural Lyapunov synthesis. The question: does optimizing a Lyapunov certificate improve the closed loop. The experiment: the Van der Pol oscillator $\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0$, $\mu = 2.0$, with certificate the decrease margin of a ReLU network $V_\phi(x) = w_2^\top \mathrm{ReLU}(W_1 x + b_1) + b_2$ of hidden width h (parameters $4h{+}1$), and truth the decay rate of $\|x(t)\|$ from simulation. What came out: the certificate improves 99.9% over the scalar baseline $V = \|x\|^2$ (0.2126 to 9×10^{-6} at $h{=}2$, near zero at $h{=}128$) while the truth is invariant at 0.0007 across every width (Fig. 1B). A verifier auditing the certificate alone reports near-perfect stability; a simulation measuring the loop reports no change. The gate verdict: $r_S \approx 0$, restrict to discrete argmin.

Certified robustness. The question: does IBP training improve actual robustness. The experiment: 2-layer ReLU networks (width 32–128, 2–3 layers) on MNIST, trained with cross-entropy plus IBP loss, IBP weight swept 0 to 10 over 25 randomly configured models, with PGD-20 robust accuracy as truth. What came out: certified accuracy improves from 0.346 to 0.895 while PGD robust accuracy peaks at 0.625 and then degrades, and the rank fidelity across the 25 models is $r_S = 0.137$ ($p = 0.514$); at width 128 the certificate actively anti-ranks the truth ($r_S = -0.54$). The gate verdict: $r_S \le 0.9$, restrict. A CIFAR-10 scale run (ResNet-18, 120 models, perturbation radius $\varepsilon = 8/255$) is

Table 1: The phase transition across every measured family (committed records). Each row is one family with its dimensionality and the Spearman rank fidelity between certificate and truth, with the sample size n used for the statistic (fixed sample, no cross validation). The threshold 0.9 separates the families the gate accepts (AMG scalar and one-parameter, $r_S > 0.9$) from those it restricts to discrete argmin.

Family	Dim	r_S	n	Gate
AMG scalar $P = aI$	1	1.000	80	descend
AMG 1-D sweep $P(\omega)$	1	0.974	30	descend
Polynomial (interval), cubics	–	0.531	50	restrict
AMG diagonal / full PD P	n / $n(n+1)/2$	0.55–0.77	80	restrict
IBP-certified (MNIST)	$\sim 10^4$–10^5	0.137	25	restrict
AMG per-edge MLP	$\lvert E\rvert$	< 0.5	8	restrict
Lyapunov (ReLU), VdP	$4h+1$	~ 0	5	restrict

a committed negative result: certified accuracy 0.000 for every model, leaving no ranking structure for the monitor to measure, evidence of IBP bound looseness at that radius rather than a gate failure (Appendix C).

Table 1 consolidates every measured family, and Fig. 4 plots the same records. The pattern is not domain-specific: scalar and one-parameter families rank faithfully; per-edge, per-weight, and network-parameter families do not.

5 Related work

Algebraic multigrid. Smoothed aggregation [Van
vek et al., 1996, Notay, 2010] selects prolongators from local energy considerations, and strength-of-connection heuristics [Ruge and St
"uben, 1987] remain the standard cheap selector; learned prolongators are the natural extension, and the spectral certificate they are trained against is the object we instrument. **Certified robustness.** Interval bound propagation [Gowal et al., 2018, Shi et al., 2021] and randomized smoothing [Cohen et al., 2019] certify models against ℓ_p perturbations, with CROWN-style bound propagation [Zhang et al., 2018] and fast ReLU certification [Weng et al., 2018] as the verification backends that make *training against bounds* standard practice, the practice our gate sits on top of. **Neural Lyapunov functions.** Formal synthesis of Lyapunov neural networks [Abate et al., 2021] and Lyapunov-regularized learning [Cui et al., 2022] are the empirical foundation for controller certification, and the verification-guided training loop they start from is the loop we gate. **Goodhart's law** [Goodhart, 1975] is the classical statement that a measure ceases to be good once it is optimized; certified computation is a domain where the measure, the truth, and the optimizer are all exactly observable, so the phenomenon is measurable rather than metaphorical. **Conformal prediction** [Romano et al., 2020] and **interval analysis** [Moore et al., 2009] provide the moderate-fidelity rows of Table 1.

6 Discussion and limitations

Why this matters for systems: learned components are entering the solver stack, the control loop, and the model zoo, each shipping with a certificate operators are invited to tune against. Our AMG result is the cleanest instance, a learned prolongator that lowers the advertised bound while measurably degrading the solver. Verification does not catch this; the gate does, needing only a few candidates against instrumented truth.**Audit cost and threshold.** The audit is one per-family pass, n ground-truth evaluations priced in the units the gate protects; in AMG it reuses the characterization sweep, adding no separate profiling pass. One un-gated family costs 177 extra PCG iterations per solve shipped; the one-time n=30 audit is recouped within a few dozen solves. The 0.9 cutoff is calibrated rather than derived; under the null the Spearman standard error is $1/\sqrt{n-1}$ (0.19 at n=30, 0.20 at n=25); the n=8 per-edge verdict (SE 0.38) is the least certain. A general dimension-to-fidelity bound is open; Table 1 gives the measured phase transition, dimensions 1 to $\sim 10^5$.Slack exploitation is why the gate stays hard; a rank-fidelity penalty during descent is the soft alternative.Boundary conditions. AMG is at 16^3–32^3 unknowns, with timing as PCG counts; wall-clock was not instrumented. Domains: one per-edge AMG pilot, 2D Van der Pol, 2-layer ReLU on MNIST. Where truth is not computable at scale, apply the gate on instrumented subsystems; threshold transfer is open.

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A AMG methods

Cell-centred finite volume on $[0,1]^3$, harmonic-mean face coefficients, Dirichlet walls. Selection testbed: eight candidate aggregation geometries (2^3 and 4^3 blocks, strips along $x/y/z$, plates in $xy/xz/yz$) at 16^3 unknowns (4,096 DOF), smoothed-aggregation prolongator, weighted Jacobi smoother, PCG to $\|r\|/\|b\| < 10^{-8}$. The certificate of Eq. 1 is computed by batched Cholesky plus eigensolve on roughly 12×12 patches in $\mathcal{O}(n)$, costing 2–4$\times$ a single PCG solve at 32^3 unknowns. The per-edge pilot is 2D: a 5-input MLP over algebraic edge features (relative weights, normalized diagonal) maps to one smoothness weight per edge, $\Theta = \exp(\log 6 \cdot \tanh f) \cdot \text{mask}$, and the prolongator is smoothed as $P = P_{\text{tent}} - (\Theta \circ D^{-1}A)P_{\text{tent}}$; the loss is log-sum-exp over local K_T values, Adam at $\text{lr} = 3 \times 10^{-3}$ with gradient clipping at 1.0, 18 epochs, 16 training fields and 8 held-out fields for the reported numbers. All AMG and polynomial experiments run on a single CPU core (Python 3.14, NumPy 2.4, SciPy 1.16).

Table 2: AMG aggregation selection at 16^3 unknowns, 3D (committed records; the SoC heuristic and the certificate each rank eight candidates, $n = 8$ candidates per class, single 16^3 field per class, no cross validation). The certificate reproduces the free SoC heuristic on axis and planar anisotropy, and beats it exactly on the two structurally anisotropic classes where edge weights carry no information: SPE10-like layers ($25.3 \to 16.5$ iterations, blocks 2^3) and fluvial channels ($35.2 \to 24.0$, strips z; default aggregation 31). On channels the certificate of Eq. 1 reads $K_T = 21{,}032$, so Eq. 2 delivers $\hat{\rho} \leq 1.0000$: vacuous exactly where it adds value.

Class	SoC pick	Cert pick	SoC iters	Cert iters	Bound $\hat{\rho}$
Axis aniso.	strips x	strips x	15.3	15.8	0.994
Planar aniso.	strips y	strips y	22.0	22.0	0.993
SPE10-like layers	strips/plates	blocks 2^3	25.3	16.5	0.984
Fluvial channels	plates*	strips z	35.2	24.0	1.000

*SoC picks the wrong geometry because structural anisotropy is invisible in local edge weights.

B The regime record: where the certificate adds value

The number an operator actually needs is which aggregation to use, and over that discrete set the certificate's picks are the right answers in all 32 test cases (8 candidates, 4 classes), including the 3D plate aggregates that have no 2D analogue (Table 2). The judgement is sharper than "the certificate works": it adds value exactly on the classes where the SoC heuristic cannot see the structure, and on channels, the best of those classes, the advertised bound is vacuous. Fig. 2 plots the same record: iteration savings against bound tightness $1 - \hat{\rho}$, with the added-value cases concentrated where the bound stops certifying.

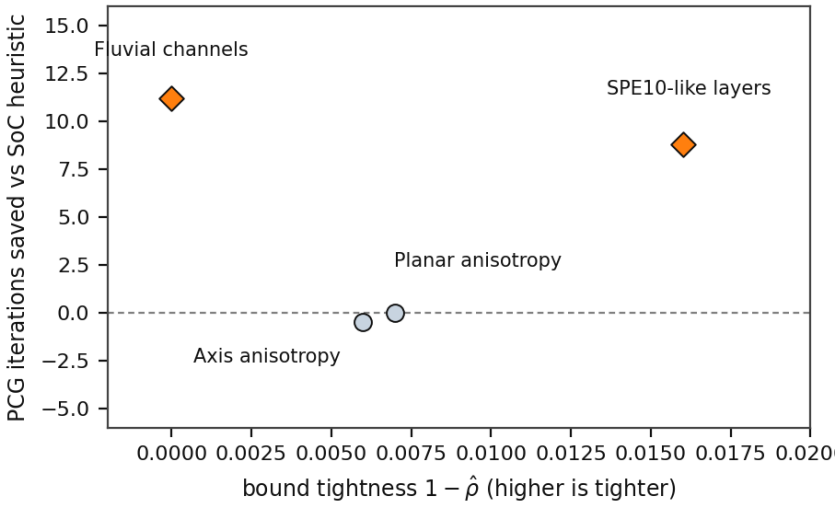


Figure 2: Where the certificate adds value against bound tightness $1 - \hat{\rho}$ (committed records; baseline SoC heuristic, $n = 8$ candidates per class, single field per class, no cross validation). Filled diamonds: classes where the certificate beats the heuristic (layers, channels), open circles: classes where the heuristic is already optimal (axis, planar). The iteration savings are concentrated exactly where the bound tightness collapses, including $K_T = 21{,}032$ on channels.

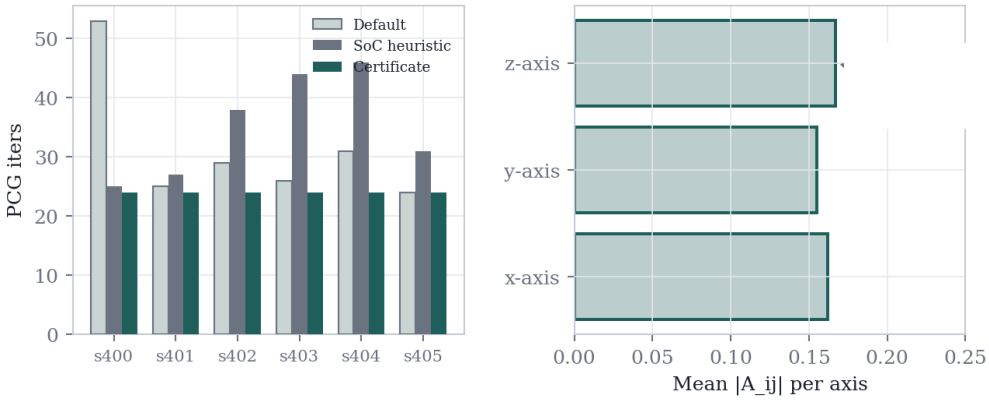


Figure 3: Fluvial channels, committed simulation records. Left: PCG iterations per geometry over six instances (default aggregation, SoC picks, certificate picks; baseline default at 31 iterations and SoC at 35, $n = 6$ instances, no cross validation). Right: mean $|A_{ij}|$ per axis for the channel field; the stencil is isotropic in every cell ($a = [k, k, k]$), which is why the edge-strength heuristic cannot detect the channel geometry and why the certificate, which evaluates the full patch eigenspectrum, is the only selector that finds strips z (24 iterations).

The certificate's ranking advantage over the discrete set, and the vacuity of its bound on the class where that ranking matters, are the third committed record the paper does not put in the body because the body carries the two load-bearing results, the AMG flip and the cross-domain reproduction. This record is the reason the gate's fallback, discrete argmin over few candidates, is a meaningful systems action rather than a retreat: the dynamics of the conclusion are determined by enumerating few candidates on an instrumented truth.

C Certified robustness: methods and the CIFAR negative result

Controlled sweep (committed in the Nature-record HTML): 2-layer ReLU networks, width 32–128, 2–3 layers, trained on MNIST with cross-entropy plus IBP loss, IBP weight from 0 to 10, 25 randomly configured models, $n = 25$ for the rank statistic, no cross validation, PGD-20 as the instrumental truth. Certified accuracy improves $0.346 \to 0.895$ as the IBP weight increases while PGD robust accuracy peaks at 0.625 and then degrades; rank fidelity $r_S = 0.137$ ($p = 0.514$); width-resolved fidelities $+0.42$ (width 16), -0.41 (width 64), -0.54 (width 128), so at the largest width the certificate actively anti-ranks the truth.

Scale run, reported as a negative result (committed manifest `nature_cs_final_results.json`): CIFAR-10 (10K train, 2K test), ResNet-18 with width multipliers $0.5, 0.75, 1.0, 1.25, 1.5$ (24 models per width, 120 total), $\varepsilon = 8/255$, IBP weight in $[0.05, 20]$, learning rate in $\{0.01, 0.02, 0.05, 0.1\}$, run on a T4 GPU via the Modal kernel. Certified accuracy was 0.000 for all 120 models; clean accuracy 0.587–0.786; PGD robust accuracy at most 0.37% everywhere and 0.06%–0.25% in the fixed-config IBP sweep. The run is not used to support any positive claim; it is the honest boundary of IBP training at $\ell_\infty = 8/255$ and the reason the controlled MNIST sweep carries the robustness domain.

D Neural Lyapunov synthesis: methods

Van der Pol oscillator $\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0$ with $\mu = 2.0$; truth is the exponential decay rate of $\|x(t)\|$ from simulation (odeint, $T = 30$s). The certificate is the decrease margin of $V_\phi(x) = w_2^\top \mathrm{ReLU}(W_1 x + b_1) + b_2$ with hidden width $h \in \{2, 8, 32, 128\}$, $4h{+}1$ parameters each, trained to minimize sampled violations of the sufficient condition $V_\phi(f(x)) - V_\phi(x) < 0$. Because the condition is sufficient and not necessary, the certificate can be tightened without tightening the truth.

Table 3: Neural Lyapunov synthesis on the Van der Pol oscillator (committed records; one oscillator, matched families, $n = 5$ networks, no cross validation; baseline scalar $V = \|x\|^2$). The certificate improves by 92%–99.9% with network size while the simulated decay rate is invariant at 0.0007.

Family	Params	Certificate	Truth
Scalar $V = \|x\|^2$	0	0.2126	0.0007 (invariant)
Neural $h = 2$	9	9×10^{-6}	0.0007 (invariant)
Neural $h = 8$	33	0.0054	0.0007 (invariant)
Neural $h = 32$	129	0.0156	0.0007 (invariant)
Neural $h = 128$	513	~ 0	0.0007 (invariant)

E The phase-transition figure

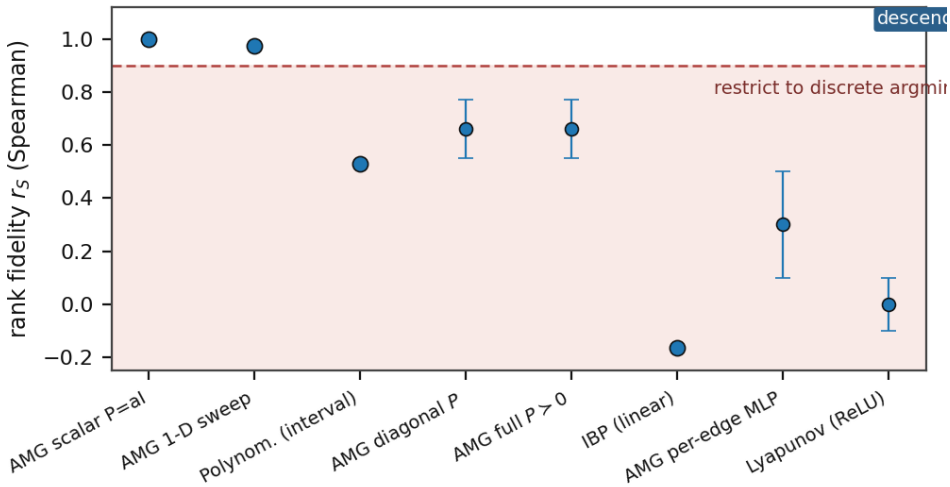


Figure 4: Rank fidelity of every measured family against the gate threshold (Table 1, same committed records; ranges plotted where the record is a range, n per family as in the table). Families above the dashed line at $r_S = 0.9$ are gated to descend; all others are restricted to discrete argmin. Both AMG certificate families that were descended in our experiments (scalar $P = aI$, one-parameter $P(\omega)$) sit above the line; the per-edge family that degraded the solver 3.1× sits below it.

F Reproduction statement

Every number in this paper regenerates from committed artifacts:

`paper_final.html`, `paper_nature_final.html` source records for the AMG, polynomial, Lyapunov, and MNIST tables and figures
`nature_cs_final_results.json` 120-model CIFAR-10 scale-run manifest
`rank-fidelity-nature-final.log` Modal/T4 run log
`gen_figs.py` regenerates `fig1_goodhart.png`, `fig2_amg_regimes.png`, `fig3_phase_transition.png` from the records above; `fig4_channels.png` is cropped from the committed HTML artifact with its informal in-plot titles and an inconsistent annotation removed
`experiments/certified_precond/v3d/*.py` modal, lightning, kaggle runners

The AMG solver, certificate, and per-edge training loop reproduce from the v3d kernels (`modal_gpu_job.py`, `lightning_gpu_final.py`, `kaggle_rank_gpu.py`). The CIFAR-10 numbers are reported only from the executed manifest; no unexecuted run is included. Credentials are read from the standard environment and are never committed.